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Air Quality and Respiratory Health: An Analysis on Healthcare System Loading

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Report for
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1 Executive summary

This report investigates the relationship between air pollution and hospital admissions between 2020 and 2025. The analysis compared the three most polluted and three least polluted cities using AQI, PM2.5, and hospital admission data. Regression analysis showed that PM2.5 was the only statistically significant pollutant associated with hospital admissions. Overall, the findings suggest that higher PM2.5 concentrations leading to higher AQI, were linked to increased hospital burden across the selected cities.

2 Introduction

Air pollution is a major public health issue causing respiratory problems leading to increased hospital admissions. Pollutants such as PM2.5, PM10, NO2, and O3 can negatively affect human health, especially in high concentrations, which are commonly found in urbanised cities. Among these pollutants, PM2.5 is considered particularly harmful because it can penetrate deep into the lungs.

This report investigates how air pollution has affected hospital admissions between 2020 and 2025, using air quality and healthcare data from eight cities. Cities were ranked based on average AQI levels, allowing comparison between the **three most polluted** and **three least polluted** cities in the dataset.

The study used descriptive analysis, visualisations, and regression modelling to examine whether higher pollution levels were associated with increased hospital burden, with a *particular focus on PM2.5 concentrations*.

3 Methodology

Question to investigate: How has air pollution affected hospital intakes in the most polluted cities in years 2020-2025?

The analysis was conducted using R and several packages from the tidyverse ecosystem for data cleaning, transformation, modelling, and visualisation. Libraries were used for date handling, plotting, table formatting, and regression analysis.

The dataset contained daily air quality and healthcare observations from eight cities between 2020 and 2025. Key variables included Air Quality Index (AQI), PM2.5, PM10, NO2, O3, temperature, humidity, hospital admissions, and hospital capacity. Daily observations were aggregated into monthly

summaries and then yearly summaries to reduce short-term fluctuations and improve trend interpretation.

Cities were ranked according to average AQI values from 2020–2025. The three most polluted and three least polluted cities were selected for comparison.

A multiple linear regression model was used to examine the relationship between air pollutants and hospital admissions. Hospital admissions were treated as the dependent variable, while AQI, PM_{2.5}, PM₁₀, NO₂, and O₃ were used as predictor variables. The regression results shown in Table 1 indicate that PM_{2.5} was the only statistically significant predictor of hospital admissions used further for analysis.

Table 1: *Linear Regression Results for Hospital Admissions*

Variable	Coefficient	Standard Error	t value	p value
(Intercept)	4.6259	0.0690	67.0640	0.0000
aqi	0.0000	0.0001	-0.4958	0.6201
pm2_5	0.0987	0.0008	126.8675	0.0000
pm10	0.0001	0.0006	0.1871	0.8516
no2	-0.0002	0.0012	-0.2097	0.8339
o3	-0.0008	0.0010	-0.8814	0.3781

Table 1 shows the regression results for the model examining the relationship between air pollutants and hospital admissions. The table indicates that PM_{2.5} is the only statistically significant predictor of hospital admissions ($p < 0.05$), while AQI, PM₁₀, NO₂, and O₃ do not show a significant association with hospital admissions in this model. As such, only PM_{2.5} was used for further analysis and visualisation to explore its relationship with hospital admissions.

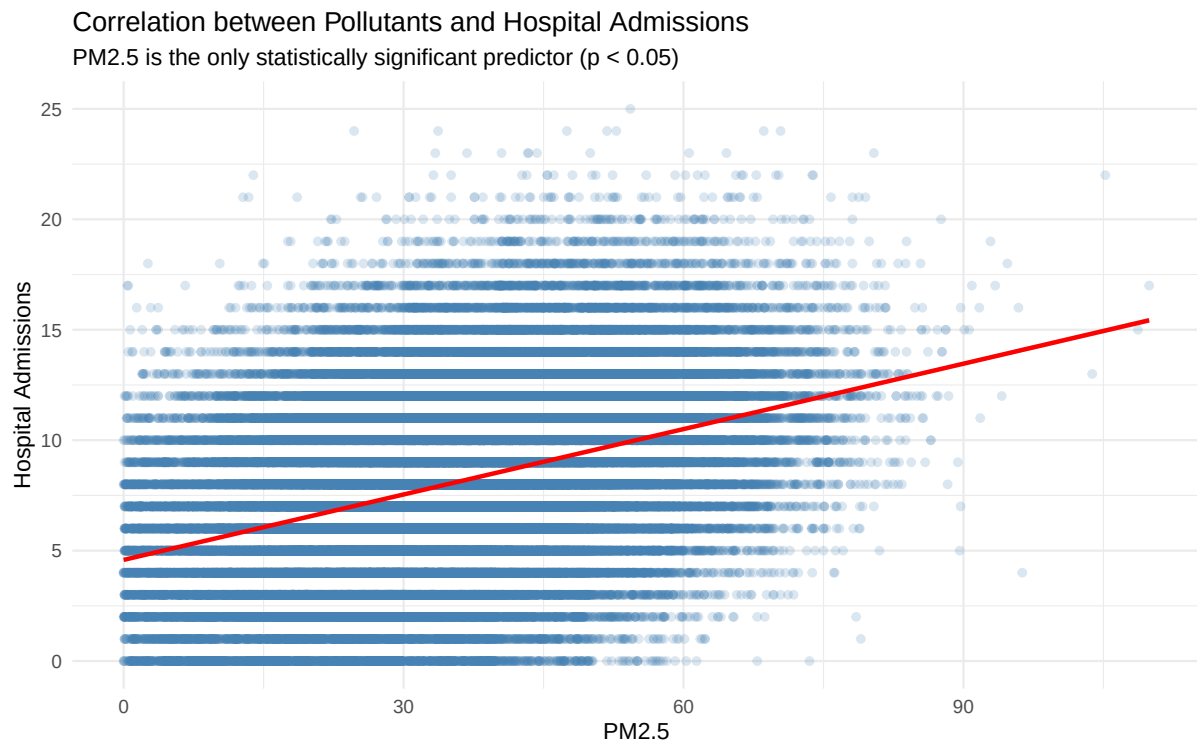


Figure 1: *Correlation between Pollutants and Hospital Admissions*

Figure 1 presents a scatterplot and fitted regression line showing the relationship between PM2.5 concentration and hospital admissions. The positive slope of the regression line suggests that *hospital admissions tended to increase as PM2.5 concentrations increased*, supporting the relationship identified in the regression model.

4 Potential Results

With help of Figure 2 we have arrived to the following potential results: * The top 3 most polluted cities generally recorded higher PM2.5 concentrations and greater hospital admissions compared with the least polluted cities.

Note: A noticeable decline in PM2.5 levels was observed from 2020 to 2021 across most cities. This was likely influenced by COVID-19 lockdowns and global shutdown measures, which reduced vehicle usage, industrial activity, and overall emissions.

- Beijing consistently showed the highest hospital burden, with hospital admissions increasing to almost 900 admissions in 2025.
- Cairo recorded large fluctuations in pollution levels and experienced the highest AQI level overall during 2024.

- Among the least polluted cities, London and Mexico City showed relatively lower and more stable pollution levels, although moderate hospital admissions were still observed.
- São Paulo recorded the highest PM2.5 concentration in 2020 despite being classified among the least polluted cities based on average AQI. This reflects that other factors might have caused the city to not follow the general pattern.

Comparison of Pollution and Hospital Admissions

Top 3 Most vs 3 Least Polluted Cities (2020–2025)

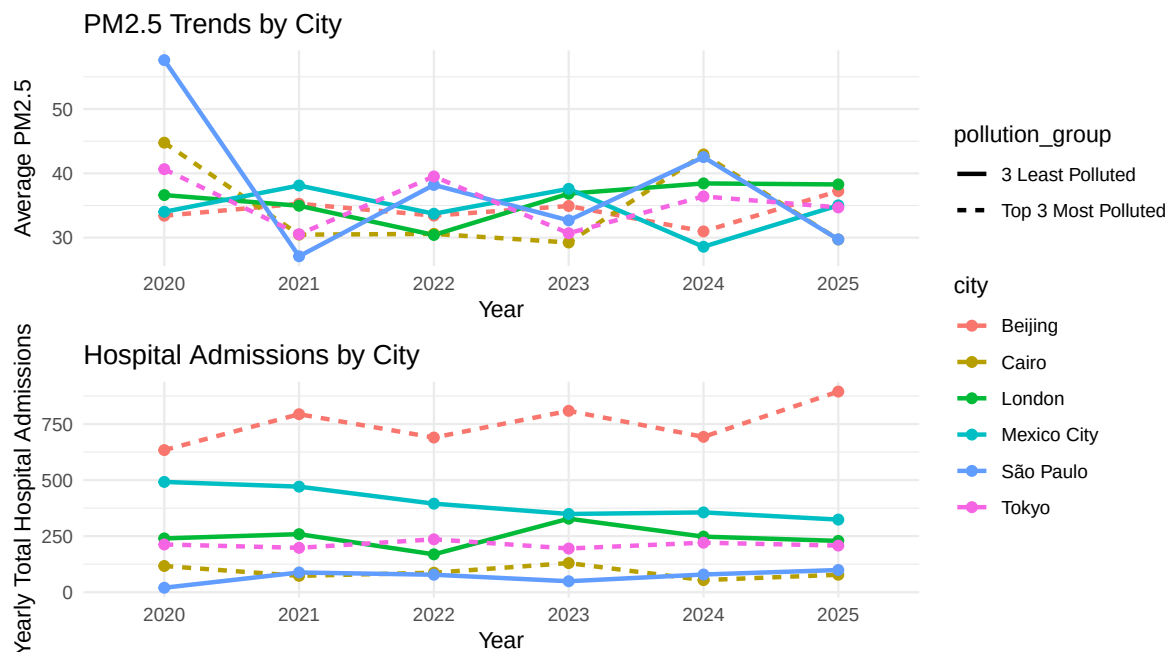


Figure 2: Comparison of Pollution and Hospital Admissions Between the Top 3 Most Vs 3 Least Polluted Cities (2020–2025)

- Table 2 identifies the most polluted year for each selected city based on AQI and PM2.5 concentrations. Cairo recorded the highest AQI value overall (417.60) in 2024, while São Paulo recorded the highest PM2.5 concentration (57.60) in 2020.
- Figure Figure 1 further supports this relationship by showing a positive trend between PM2.5 concentration and hospital admissions.

Table 2: Most Polluted Year for the Top 3 and Least 3 Polluted Cities (2020–2025)

City	Most Polluted Year	Average AQI	Average PM2.5
Beijing	2025	289.64	37.24
Cairo	2024	417.60	42.94
London	2025	276.33	38.27

Mexico City	2023	264.17	37.59
São Paulo	2020	392.00	57.60
Tokyo	2021	305.62	30.53

Overall, the findings suggest that increasing PM2.5 concentrations were associated with increased hospital burden between 2020 and 2025.

5 Discussion

The analysis indicates a positive relationship between air pollution and hospital admissions, particularly for PM2.5 concentrations. Cities with higher pollution levels generally recorded greater hospital burden, supporting existing research linking particulate pollution with respiratory health issues.

The regression model further identified PM2.5 as the strongest predictor of hospital admissions among all selected pollutants.

Several limitations should be acknowledged:

First, the dataset only included a limited number of cities (8), reducing the ability to generalise findings globally. The data does not carry data previous to 2020 which makes it difficult to give a more broad look at the investigating question.

Second, hospital admissions may also be influenced by factors not included in the analysis, such as population density, healthcare accessibility, socioeconomic conditions, smoking prevalence, and public health policies.

Third, yearly aggregation may hide short-term pollution spikes and seasonal health impacts. Fourth, the dataset has been taken at the courtesy of the *Kaggle* platform and hence cannot be relied to make any real-world conclusions or predictive analysis, based on the findings. It has been used solely for the purpose of teaching and understanding.

In addition, cities classified as “least polluted” are according to the dataset and not the current global scenarios. However their high values during certain years, thus indicate that pollution remained a concern across most locations in the dataset. The analysis also focused on correlation rather than direct causation.

6 Conclusion

This study examined the relationship between air pollution and hospital admissions across selected cities between 2020 and 2025. The findings demonstrated that cities with higher pollution levels generally experienced greater hospital burden.

PM2.5 was identified as the most statistically significant pollutant associated with hospital admissions. Both the regression model and visualisations showed a positive relationship between PM2.5 concentration and healthcare demand.

7 Recommendations

On the basis of the above study and analysis on the dataset. Here are the following recommendation from consultant 6it

- Governments should strengthen environmental regulations aimed at reducing PM2.5 emissions from traffic, industrial activities, and urban pollution sources.
- Healthcare systems in highly polluted cities should improve hospital preparedness and respiratory healthcare capacity.
- Future studies should include additional variables such as population density, healthcare access, and socioeconomic conditions to improve analysis accuracy.
- Future research using daily or seasonal data may provide deeper insight into short-term pollution impacts on respiratory health.
- Public awareness campaigns should encourage protective measures during periods of poor air quality.